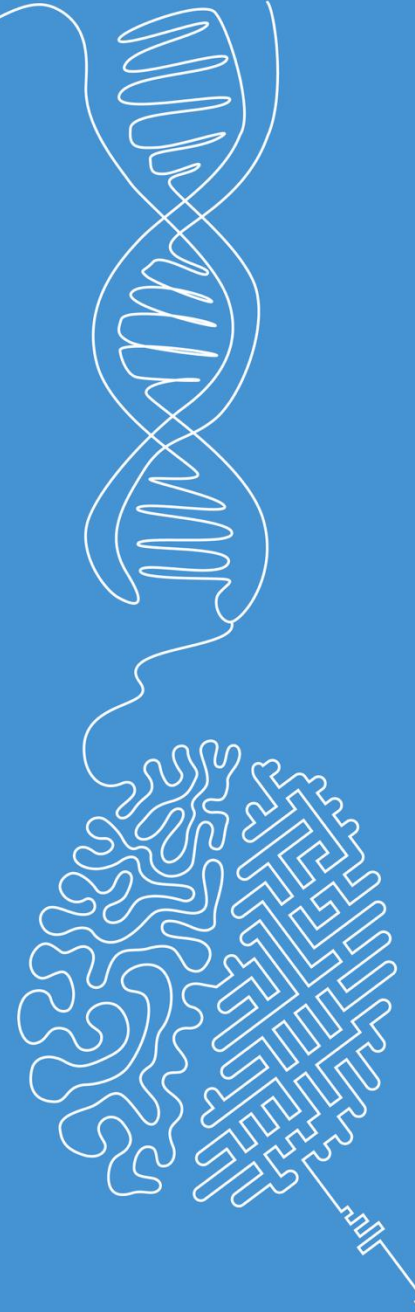


Quick response code

Image and Signal Processing

Norman Juchler



QR codes

An image processing application

Quick response code: QR code

- 2D barcode invented in 1994
- Inspired by the Japanese board game Go
- Designed for fast and flexible data encoding
- Encodes numeric, alphanumeric, and binary data
- Stores up to ~3'000 characters (version-dependent)
- Provides built-in error correction
- Readable by most smartphone cameras
- Easily generated using tools and libraries:
 - Python: [qrcode](#)
 - Web: [Calm9](#) (select QR code version)
 - Web: [Adobe Express](#)
- ISO standard: [ISO/IEC 18004](#)



Exemplary QR codes.
Top: version 2 (21x21 pixels) Bottom:
version 4 (33x33 pixels). Guess where
the codes point to...

QR code: Structure

QR codes all follow a defined structural pattern (which varies slightly across versions). This illustration provides an overview of the key components that enable reliable detection, orientation, and robust data decoding.

**Positioning / Orientation**

Required for the QR code to be detected and localized

**Format information**

Stores data about the error correction level and mask pattern

**Timing marks**

Used to define the pixel grid

**Version information**

QR code version: 1...40

**Spacing**

For better detectability

**Alignment pattern**

Used to correct perspective skew and other distortions in the image.

QR code: Error correction

- Built-in error correction allows reading even if the code is partially damaged
- Core ideas:
 - Add extra, redundant data in the form of mathematical patterns (Reed-Solomon codes)
 - This enables the QR scanner to detect and correct errors in the data
 - Depending on the correction level, up to ~30% of the code can be recovered
- Note: The method cannot recover from all errors, as demonstrated [here](#).

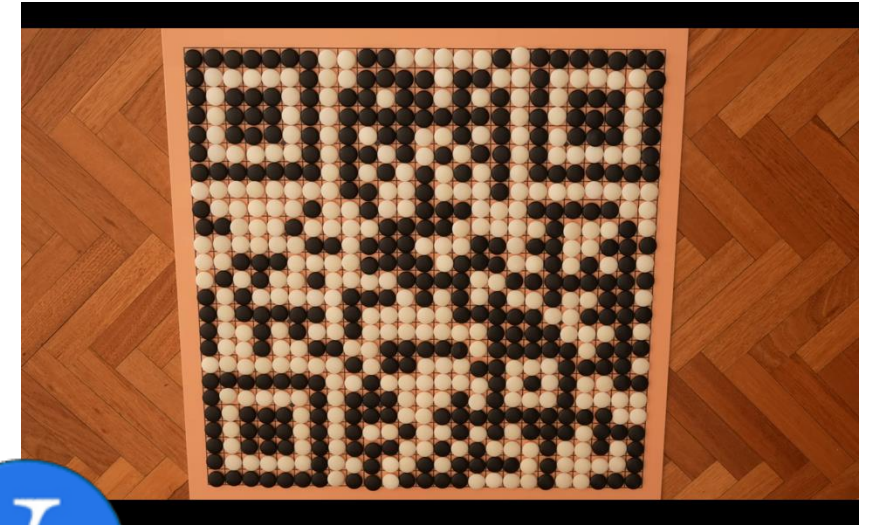
Example of a damaged QR code – a scanner can still read the encoded information thanks to error correction.
Source: [Wikimedia](#)



QR code

- From Morse code to barcodes to QR codes...
- An excellent explanatory video by Derek Muller / Veritasium
- [Link](#)

You**Tube**



I built a QR code with my bare hands to see how it works

Thanks to Yassin Rhouma / AD22 for the valuable suggestion to include this topic 😊